Artificial NEURAL NETWORKS (ANN)

1. What is an Artificial Neural Network?

An Artificial Neural Network (ANN) is a **machine learning model** inspired by how neurons work in hom human brain.

Human Brain:

- Neurons processs information via electrical signals.
- They connect through synapses and "fire" when activated.
- ANN: Artificial **neurons** are mathematical functions.
- They connect via **weights** and process inputs to

2. ANN Architecture

Layer	Role
Input Layer	Accepts features of the dataset (e.g.. pixels of an image).
Hidden Layer(s)	Performs complex computations and feature extraction. Can be mulfiple.
Output Layer	Provides the final result

Each Neuron:

- Receives multiple inputs.
- Applies **weights** to inputs.
- Adds a bias.
- Passes result through an **activation** function

Neuron Mathematical Formula

$$z = w_1 x_1 + w_2 x_2 + w_n x_n + b$$

$$a = \text{Activation}(z)$$

Activation Functions

Function	Formula	Use Case
Sigmoid	$\frac{1}{i+e} e^{e}$	Binary classification
Tanh	$\tanh(z) \circ \frac{e^{x} - e^{z}}{z^{x} + e^{x}}$	Output in [–1, 1]
ReLU	$\max(0, z)$	Fast learning: common in deep networks
Leaky ReLU	x if $x > 0$, else $0{,}01x$	Solves "dead neuron" problem in ReLU
Softmax	x if $x > 0$, else $0{,}01x$	Used in output layer of multi-class classification

Forward Propagation

- Step-by-step data flow:

1. Input is multiplied with weights prediction is from actual value.
2. Output passed througl. activation func.
3. Output of one layer = input to next.
4. Final output generated

Loss Function

Measures how far the network's prediction is from the actual value.

Common Loss Functions:

- Mean-Squared Error (MSE) $\frac{1}{n}\sum(y-\bar{y})^2$
- Cross-Entropy Loss (for classification)

Backpropagaation + Gradient Descent

Backpropagation = How the network learns.

1. Calculate loss/error at output.
2. Compute gradients (how much each weight contributed to error.)
3. Update weights using Gradient Descent:

Hyperparamenters to Tune

$$w = w - \alpha \frac{\partial Loss}{\partial w}$$

where α is the learning rate.

9. Hyperparameters to Tune

Hyperparam.	Description
Learning Rate	Size of each step during weight updates
Epochs	Number of times entire dataset is passed through
Batch Size	Number of somples processed before updating the model
Hidden Layers	Number/depth of hidden layers
Neurons per layer	Affects model complexity and performance

9. Example Use Case: Handwritten Digit Recognition (MNIST)

- Input: 28 × 28 grayscale image → 784 inputs
- Hidden Layers: Use ReLu
- Output: 10 neurons → Each neuron corresponds to digit 0–9

Output Activation: Softmax
Loss Function: Cross-Entropy

Types of Neural Networks (Extensions of ANN)

Type	Description
Feedforward Neural Network (FNN)	Basic ANN; data moves forward only
Convolutional Neural Network (RNN)	For images
Recurrent Neural Network (DNN)	For sequences (like time series or text)

Real-Life Applications

- Face Recognition
- Speech-to-Text (Google Assistant, Siri)
- Self-driving Cars (Image + Sensor analysis)
- Medical Diagnosis (X-ray, ECG classification
- Stock Market Prediction